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Manuscript

Urban Analytics and City Science

EPB: Urban Analytics and City Science
2026, Vol. 0(0) 1–24
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DOI: 10.1177/23998083261488638
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Spatial reach of visual information: Optimizing buffer distances for computer vision–based measures of microscale walkability and visual quality

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Abstract

An increasing number of studies use computer vision and street view images to evaluate how favorable a location is for pedestrians. Yet the spatial reach of visual information that informs pedestrian perceptions has received little systematic investigation, resulting in widely varying buffer distances across walkability studies. Furthermore, studies using these image-based metrics often overlook the possibility that optimal buffer distances may vary across different trip purposes and pedestrian characteristics (e.g., older adults), further exacerbating the uncertain geographic context problem of visual information. This paper examines the spatial reach of image-based walkability indicators by identifying the buffer distances at which image-based microscale and visual quality measures best explain walking environment satisfaction in Seoul, Korea. Walking environment satisfaction was measured through intercept surveys in which pedestrians rated their satisfaction with the surrounding environment for walking on a five-point scale. Findings show that optimal buffer distances for image-based microscale features generally range between 100m and 400m. For visual quality indicators, optimal buffers are 200m or less for most groups, which is significantly smaller than 1000-1400m distances found for GIS-based macroscale factors. Trip purposes and familiarity of the area result in distinctive patterns in contributing walkability factors and their optimal buffer distances. Visual quality measures offer substantial improvements to model fits if appropriate buffer distance is used. A higher macroscale index, greater building-to-street ratio, and safer-looking locations are associated with higher walking environment satisfaction.

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Data Availability Statement included at the end of the article

Livelier-looking locations showed negative relationship. Implications for policy and research are discussed.

Keywords
walkability, spatial reach of visual information, pedestrian characteristics, street view images, computer vision

Introduction

Walking is the most basic way to navigate in a city. How favorable a given location is for walking is shaped by various environmental factors, including macroscale factors (e.g., connectivity and proximity to destinations), microscale urban design details, and visual quality (Alfonzo, 2005; Koo et al., 2021; Saelens et al., 2003). Recent studies increasingly use computer vision and street view images to quantify microscale features and visual quality—two aspects of urban environment previously considered 'unmeasurable' (Ewing and Handy, 2009; Yang et al., 2019; Yao et al., 2019). Numerous studies have now established the effectiveness of image-based measures of microscale and visual quality in capturing walkable streetscapes. These measures commonly operationalize walkability of a location using the characteristics of its *surrounding environment*, typically by drawing buffers around locations of interest to characterize environmental characteristics. For these buffer-based studies, identifying the actual area that "exert contextual influences on the individual being studied" is a paramount task because failing to do so may be "a major reason why research findings concerning the effects of social and physical environments on health behaviors and outcomes are often inconsistent" (Kwan, 2012, p. 959).

Despite the rapid growth and repeated demonstration of the effectiveness, a persistent methodological limitation in measuring microscale features and visual quality using computer vision is *the lack of clear guideline on the configuration of the "surrounding environment" for image-based measures of microscale and visual quality indicators* of walkable streetscapes. Before computer vision, the traditional methods of measuring urban design details in microscale and visual quality have been surveys. Surveys typically define the relevant spatial extent by asking respondents to evaluate their "immediate neighborhoods", "within easy walking distance", or "10-15 minutes of walking" (Adams et al., 2009; Sallis et al., 2010).

Unlike surveys, however, objectively measuring microscale features and visual quality does not involve human respondents who can indicate their perceived contextual units and therefore researchers must make that choice, often arbitrarily and without clear theoretical underpinnings. For example, studies employing researcher-defined buffers for microscale features have used inconsistent buffer distances ranging from 150m (Koo et al., 2021), 500m (Ki and Lee, 2021), 800m (Yang et al., 2019), to 1000m (Lu, 2018; Nagata et al., 2020) without providing theoretical or empirical reasons for the choice, suggesting a potential for the uncertain geographic context problem (Kwan, 2012, p. 959).

While many researchers have examined how different buffer configurations influence walkability metrics, their primary focus has been on macroscale factors, and systematic investigations on the influence of buffer distances on microscale and visual quality have been limited. For example, Lam et al. (2022) measured walkability using population density, retail and service density, land use mix, street connectivity, green space, sidewalk density, and public transport density at various buffer distances (150, 500, and 1000 m), with 150m providing the largest effect sizes. Christiansen et al. (2016) examined 500 m and 1 km buffers and reported that residential density and land use mix generally showed stronger associations with walking for transport across

multiple cities when measured using a 500 m buffer. James et al. (2014) tested 400, 800, 1200, and 1600 m buffers and found 1200 m buffers to show the largest effect sizes. Note that the findings of these macroscale-focused studies yielded inconsistencies about optimal buffer distances. Some studies reported little influence of different buffer distances on the fit of the statistical models tested (Forsyth et al., 2012; Villanueva et al., 2014).

This inconsistency is not limited to residence-based neighborhood studies. Walkability studies that focused on non-residential locations often rely on arbitrarily chosen buffer distances that vary substantially, ranging from 50m to 1600m—such as various origins or bouts of trips that took place throughout a day (Kang et al., 2017; Koo et al., 2023), sample of urban streets (Yin and Wang, 2016), survey sites (Kim et al., 2014; Lee et al., 2020; Tamura et al., 2019; Troped et al., 2010), and urban intersections (Miranda-Moreno et al., 2011; Park et al., 2023; Pulugurtha and Sambhara, 2011; Schneider et al., 2009). Some studies adopted the same buffer distance for macroscale and microscale environmental characteristics, implicitly assuming that the 'causally relevant geographic unit' are identical for both scales—an assumption that, to our knowledge, has not been empirically tested for image-based measures (Ki and Lee, 2021; Lu, 2018; Yang et al., 2019).

One potential reason that past studies used widely varying buffer distances may be that the appropriate sizes may vary depending on pedestrian characteristics and trip purposes. Many existing studies tended to use a "one-size-fits-all" approach that focuses on general populations rather than specific groups or settings (Manaugh and El-Geneidy, 2011; Stafford and Baldwin, 2018). Recognizing that different trip purposes and demographic groups may have distinct preferences and needs, walkability measurement methods should be customized for each group and context. For example, Stafford and Baldwin (2018) noted that conventional walkability measurement methods–which are centered on standardized norms related to physical ability and space–may not work well for certain socio-demographic groups (e.g., older adults). In this context, Villanueva et al. (2014) identified measurable differences in optimal buffer distances according to life stages, underscoring the need for tailored approaches.

Additionally, most studies examining the impact of different buffer distances have focused on populations in North America and Europe (Shields et al., 2023). Because the buffer distances reflect the typical walking distances of the specific population of interest, which may vary across different cultures and cities, the applicability of buffer distances across different cities remains underexplored.

In summary, several research gaps persist in the literature:

(1) The appropriate buffer configuration for image-based measures of microscale features and visual quality of streetscapes, and how they differ from those for the traditional macroscale walkability factors, remain largely untested.
(2) There is limited investigation into how these optimal buffer distances vary across different demographic groups or trip purposes.
(3) Most studies have been conducted in North American and European settings, leaving the relevance of these findings to many other regions in the world unverified.

This study aims to address these gaps by (1) examining optimal spatial buffer distances for macroscale, microscale, and visual quality measures in explaining walking environment satisfaction and (2) exploring how the optimal buffer distances vary across different population groups and trip purposes using pedestrian survey data from Seoul, Korea. It is important to reiterate that this study focuses on walking environment satisfaction measurements in the pedestrian survey—a subjective evaluation of the perceived quality of the walking environment at a specific location—which is conceptually distinct from observed walking behavior such as trip frequency, duration, or mode choice. While the two constructs can be correlated, they capture different dimensions of

walkability: the former reflects how people experience and evaluate streetscapes, whereas the latter reflects realized travel choices shaped by additional factors such as trip necessity, time constraints, and alternative mode availability.

Data and methods

Walking environment satisfaction from Seoul survey

In September and October of 2013 and 2014, the Seoul Metropolitan Government surveyed pedestrians at 1000 locations each year across the city. The survey locations were intended to represent diverse pedestrian environments across Seoul and were selected based on various characteristics such as land use, public transit availability, and streetscape conditions, including sidewalks (Seoul Metropolitan Government, 2013, 2014).

At each of the 1000 locations, 20 intercept surveys of pedestrians were performed by trained surveyors, yielding a total of 20000 responses per year. Specifically, 10 respondents were surveyed on Friday and 10 on Saturday at each site, allowing the survey to capture both weekday and weekend contexts. The target population consisted of male and female pedestrians aged 15 or older passing through the selected locations. While pedestrian trips may involve multiple purposes, the survey recorded only the respondent's primary trip purpose.

Pedestrians were asked to score their satisfaction with the surrounding environment for walking using a five-point scale from 1 (very dissatisfied) to 5 (very satisfied). Pedestrians were also asked about their demographic and trip characteristics (Table 3). The responses were largely symmetrical, with skewness ranging between −0.012 and 0.182 across the demographic and trip purpose groups (See Appendix A for details). The individual responses were the unit of analysis of this study.

We excluded incomplete responses and locations with less than 10 street view images within 100 m buffers from the analysis. This process resulted in a dataset of 15200 valid samples from 760 locations in 2013 and 14640 valid samples from 732 locations in 2014. Because 29 locations were observed in both years, the final analytic sample comprised 29,840 responses nested within 1,463 unique locations.

Measuring macroscale factors

The macroscale index was adopted from Frank et al. (2005), which has become one of the most widely used measures in walkability research and has been extensively validated in subsequent studies (e.g., Venerandi et al., 2024; Weng et al., 2019). The macroscale factors were measured using residential density, intersection density, and land use mix (Frank et al., 2005). Residential density was measured as the number of housing units in the Korean-equivalent of Census Blocks (known in Korea as jibgye-gu, the smallest unit used for statistical tabulation by Statistics Korea) within a given buffer divided by the area of the buffer. Intersection density was calculated by dividing the number of intersections by the area of the buffer. For the land use mix, building uses

Table 1. Prediction performance of the qualitative environmental attributes: correlation between prediction and ground truth.

Metrics	Safe*	Lively*	Beautiful	Wealthy	Depressing	Boring
Correlation between prediction and ground truth	0.66	0.57	0.59	0.51	0.57	0.36

Asterisks (*) indicate visual qualities that are included in the subsequent analysis.

were classified into one of the following: educational, entertainment, office, retail, single-residential, multi-residential, and other uses. The areas of the building footprint for each of the uses were plugged into the following equation:

$$land\ use\ mix = -1 * \sum_{i=1}^{n} \left[\frac{Area\ of\ land\ use_i}{Total\ area\ of\ all\ land\ use\ categories} * ln\left(\frac{Area\ of\ land\ use_i}{Total\ area\ of\ all\ land\ use\ categories} \right) \right] / \ln(n)$$

where n is the number of land use classes considered. These metrics were then combined into an index by using the following formula: Macroscale index = 6*z-land use mix + z-residential density + z-intersection density where z- prefix represents the z-score transformation (Frank et al., 2005). The six-fold weight given to land use mix was determined in Frank et al. (2005) as the results of empirical investigations for the greatest explanatory power for physical activity. This index has been widely used and verified in previous studies for its effectiveness in capturing walkable built environment (Venerandi et al., 2024; Weng et al., 2019). Residential density, intersection density, and land use mix were measured using data from 2015, 2014, and 2017, respectively, which align closely with the 2013-2014 walking environment satisfaction survey period (See Table 2 for the detailed data sources and definitions). Although the building-use data used to calculate land use mix involve a 3-year gap, they represent the earliest available dataset for this variable. It is also worth noting that land-use characteristics are generally relatively stable over time. These were measured using eight different buffer distances, from 200 m to 1600 m at 200 m interval. This resulted in eight

Table 2. Summary of built environment variables used in the analysis.

Type	Variable	Definition	Data source (Data year)	Buffer range tested
Dependent variable	Walking environment satisfaction	Five-point satisfaction score for the surrounding environment for walking	Seoul Metropolitan Government (2013, 2014)	-
Macroscale	Land use mix	Entropy-based index of land-use diversity	MOLIT (2017)	200-1600 m
	Intersection density	Count of intersections per unit area	Seoul Metropolitan Government (2014)	
	Residential density	Count of residential units per unit area	MOLIT (2015)	
	Macroscale index	Weighted composite of z-scored components following Frank et al. (2005)	Above three sources	
Microscale	Building-to-street ratio	(%Building + %house)/(% sidewalk + %road + %car)	Naver street view (2013, 2014) + SegFormer (ADE20K)	100-800 m
	Greenness	%Tree + %grass + %plant (log-transformed)		
Visual quality	Safety	Predicted safety score	Naver street view (2013, 2014) + CV model trained on Place Pulse 2.0	
	Liveliness	Predicted liveliness score		

Note. MOLIT = Ministry of Land, Infrastructure and Transport. CV = Computer vision.

different measurement versions, one for each buffer distance, of macroscale index per survey location.

Measuring microscale features

This study leverages Naver Street View (NSV) images to assess microscale features and visual quality in the urban environment. Naver is a Korean company that provides search engine and mapping services similar to Google. We downloaded images at every 50 m along streets within the designated buffers. The images taken in 2013 and 2014 were used to match the time frame of the survey data. To more accurately capture greenery, we excluded images taken during the winter months. As mentioned above, we excluded locations with less than 10 NSV images. The total number of images used in the analysis is 911,047.

To process the images, we employed SegFormer trained on the ADE20K dataset to perform pixel-wise classification into predefined object categories (Xie et al., 2021; Zhou et al., 2017). We created buffers of various sizes around each survey location to summarize the output from SegFormer using the following formula as described in Koo et al. (2023) and Ki and Lee (2021):

$$\text{building-to-street ratio} = (\%\ \text{building} + \%\ \text{house})/(\%\ \text{sidewalk} + \%\text{road} + \%\ \text{car})$$

$$\text{greenness} = \%\ \text{tree} + \%\ \text{grass} + \%\ \text{plant}$$

where building-to-street ratio represents the canyon-like or room-like property of streetscapes that give pedestrians sense of enclosure (Harvey et al., 2015); greenness represents the pedestrians' actual exposure to street greenery, which has been widely employed in previous studies examining the relationship between greenery and walking behavior (Ki and Lee, 2021). These metrics have been calculated for eight buffer distances from 100 m to 800 m. Greenness was log-transformed due to its skewness.

Measuring visual qualities

The Place Pulse 2.0 dataset was used to train computer vision models for the large-scale assessment of visual qualities. This dataset was created by the MIT Media Lab (Dubey et al., 2016) through crowdsourced contributions of voters who were presented with pairs of randomly selected Google Street View images from 56 cities around the world. The voters were asked to vote on one of the two images to evaluate six visual qualities including safe, lively, boring, wealthy, depressing, and beautiful (Dubey et al., 2016). The dataset contains roughly 1.6 million votes on 110,988 images. Since the introduction, this dataset, or other datasets that adopted similar framework, has been extensively used in subsequent studies and showed effectiveness in explaining streetscape perceptions or associated behavioral outcomes (De Nadai et al., 2016; Wang et al., 2019).

This pairwise comparison data on the visual quality of images was converted into a continuous scale using a rating algorithm called TrueSkill (Herbrich et al., 2006) and normalized into z-scores, resulting in a training image dataset of visual quality across the six visual qualities. We used transfer learning techniques to train six computer vision models, each predicting one of the six visual dimensions. Table 1 shows the correlation coefficient between the predicted values from the trained computer vision models and the ground truth values using a validation split (20% of the dataset that the models have not seen in the training) for the six visual qualities. Five out of the six visual qualities showed correlation coefficients on par with or greater than the benchmark score of 0.54 reported by Naik et al. (2014). These six computer

Table 3. Descriptive statistics of key variables.

Variables	Continuous variables: mean (SD) Categorical variables: Count (%)		
Level 1 variables			
n	29840		
Satisfaction (Range: 1∼5)	3.17 (0.97)		
Purpose	Utilitarian:	27403 (91.8%)	
	Recreational:	2437 (8.2%)	
Frequency of visit to the location	Less than once a month:	5680 (19.0%)	
	1-2 times/week:	5196 (17.4%)	
	3-4 times/week:	9253 (31.0%)	
	Every day:	9711 (32.5%)	
Age	15-19:	1516 (5.1%)	
	20-29:	5688 (19.1%)	
	30-39:	5341 (17.9%)	
	40-49:	5309 (17.8%)	
	50-59:	6950 (23.3%)	
	Over 60:	5036 (16.9%)	
Year	2013:	15200 (50.9%)	
	2014:	14640 (49.1%)	
Gender	Female:	16045 (53.8%)	
	Male:	13795 (46.2%)	
Occupation	Student:	4082 (13.7%)	
	Homemaker:	6240 (20.9%)	
	Executive/Managerial:	1186 (4.0%)	
	Military personnel:	11 (<0.1%)	
	Unemployed/Others:	2205 (7.4%)	
	Office/Technical worker:	4377 (14.7%)	
	Manufac./Transport. Worker:	845 (2.8%)	
	Manual worker:	1628 (5.5%)	
	Self-employed:	3327 (11.1%)	
	Professional/Freelancer:	2534 (8.5%)	
	Sales/Service worker:	3405 (11.4%)	
Level 2 variables			
N		1463	
Macro index (400m)		−0.015 (5.93)	(Unit: Weighted sum of z-scores)
Microscale	Building to street Ratio (100m)	0.72 (0.26)	(Unit: Ratios of pixel percentages)
	Greenness (100m)	1.479 (0.795)	(Unit: Sum of pixel percentages)
Visual quality	Safety (100m)	0.21 (0.43)	(Unit: Z-score)
	Liveliness (100m)	0.26 (0.36)	(Unit: Z-score)

Note. All variables were converted to z-scores before the model entry.

vision models were then applied to the downloaded NSV images and averaged within each buffer to predict their visual quality.

It is expected that many of the visual qualities are correlated with one another (F. Zhang et al., 2018). We examined mutual correlations and excluded measurements for wealthy, depressing, and boring due to strong correlation ($|r| > 0.7$). We also examined variance inflation factors (VIF) of the fitted models and excluded beauty due to VIF greater than 10. We therefore included safety and liveliness in the subsequent analysis.

Statistical analysis

Multilevel models are used to statistically model the relationship between walking environment satisfaction (dependent variable) and macroscale index, microscale features, and visual quality measures. We initially treated the dependent variable, measured using a five-point Likert scale, as an ordinal variable and developed pilot models using multilevel ordinal logit regression. However, the models failed to meet the proportional odds assumption. Instead, we used multilevel ordinary least squares regression with survey respondents at Level 1 and survey locations at Level 2, specifying random intercepts at the location level. The two-level random-intercept linear model can be expressed as follows:

Level 1 (individual i within location j)

$$\boldsymbol{Y_{ij}} = \beta_{0j} + \sum_k \boldsymbol{\beta_k X_{k,ij}} + \boldsymbol{\varepsilon_{ij}}, \boldsymbol{\varepsilon_{ij}} \sim \boldsymbol{N}\left(\boldsymbol{0}, \boldsymbol{\sigma^2}\right)$$

Level 2 (location j)

$$\beta_{0j} = \boldsymbol{\gamma_{00}} + \boldsymbol{\gamma_1} \cdot \boldsymbol{MacroIndex_j}(\boldsymbol{d_m}) + \boldsymbol{\gamma_2} \cdot \boldsymbol{Building2StreetRatio_j}(\boldsymbol{d_s}) + \boldsymbol{\gamma_3} \cdot \boldsymbol{Greenness_j}(\boldsymbol{d_s})$$

$$+\boldsymbol{\gamma_4} \cdot \boldsymbol{Safety_j}(\boldsymbol{d_v}) + \boldsymbol{\gamma_5} \cdot \boldsymbol{Liveliness_j}(\boldsymbol{d_v}) + \boldsymbol{u_j}$$

Random effects

$$\boldsymbol{u_j} \sim \boldsymbol{N}\left(\boldsymbol{0}, \boldsymbol{\tau^2}\right), \varepsilon_{ij} \sim \boldsymbol{N}\left(\boldsymbol{0}, \boldsymbol{\sigma^2}\right)$$

where, $\boldsymbol{Y_{ij}}$ is the walking environment satisfaction score of pedestrian i surveyed at location j; β_{0j} is the location-specific intercept; $X_{k,ij}$ is the k th individual-level covariate for pedestrian i surveyed at location j; β_k is the fixed effect coefficient associated with the k th individual-level covariate; $\boldsymbol{\gamma_{00}}$ is the grand intercept; $\boldsymbol{u_j}$ is the location-level random intercept capturing unobserved heterogeneity across locations; ε_{ij} is the individual-level residual; and $\boldsymbol{d_m}, \boldsymbol{d_s}, \boldsymbol{d_v}$ are the buffer radii for macroscale, microscale, and visual quality measures, respectively, each optimized independently.

Demographic and trip characteristics of survey respondents were included as the control independent variables, including sex, age, occupation, trip purpose, and visit frequency. When one of the control variables was used to define a group, it was excluded from the regression. We also added year in which the survey was done as an independent variable to control for the possibility of systematic differences between the 2 years.

This study considers the model with the best model fit as representing the optimum buffer. Marginal R-squared is used to examine the model fits. The fit of multilevel models can be assessed using two different types of R-squared metrics: marginal R-squared and conditional R-squared. Marginal R-squared represents the variance explained by the fixed effects while the conditional R-squared represents the variance explained by the entire model. We hereafter refer to the Marginal R-squared simply R^2.

The analysis of this study is conducted in three steps. Step 1 focuses on the results obtained from using only the macroscale index, which comprises the three macroscale factors: residential density, intersection density, and land use mix. This step aligns most closely with most previous walkability studies, allowing for a more direct comparison with existing literature. Step 2 expands the analysis by adding the two microscale features to models from Step 1. Finally, Step 3 adds the two visual quality measures to the models from the previous step.

Results

Descriptive statistics

Table 3 presents the descriptive statistics of the variables used in the following analyses. Of 29,840 samples, the majority (91.8%) were utilitarian walking (Utilitarian group), and only 8.2% were recreational walking (Recreational group). Frequency, defined as how often respondents visit the survey location, shows that about 81.0% were familiar with the location (i.e., visiting the location at least once per month) and about 19.0% were unfamiliar with the location (i.e., visiting the location less than once a month; Infrequent visitors group). Age groups were generally evenly distributed except for 15-19 (5.1%), followed by over 60 (16.9%; Elderly group). Year and gender (Male and Female groups), both of which have two classes, had a roughly even divide. The demographic composition of the respondents was reasonably comparable to the demographic profile of residents of Seoul. The ratio between male and female in Seoul in 2013 and 2014 was 0.97 whereas the ratio in the data used in this study was 0.86, suggesting that female respondents were somewhat over-represented. The age distribution of the data used in this study resembles closely with the demographic profile of Seoul. With the exception of 50-59 age brackets being over-represented, the bracket-wise difference is less than 3% (See Appendix B for details). The occupation distribution data of Seoul was using different categorizations, making the data incomparable to the data used in this study. Because macroscale index was constructed by summing z-scored variables, its mean and standard deviation does not convey interpretable information. Building-to-street ratio (measured with 100m buffer) had the mean of 0.72, the minimum of 0.08, and maximum of 2.30, indicating there were more road pixels than building pixels, with a few exceptionally high building coverages. Greenness—the sum of tree, plant, and grass (values exponentiated for descriptive purposes)—had a mean of 1.479 and the range of 0.059 and 37.509. This indicates Seoul had limited street greenery on average, with a few highly green streets. The two visual quality measures had similar means, 0.21 and 0.26 for safety and liveliness, respectively. Since these variables were in z-scores, safety and liveliness of Seoul's sample streets are slightly safer and lively-looking than the average of the 56 cities in Place Pulse 2.0 dataset.

Step 1: buffer configuration for macroscale factors

We first present analysis results under the setting most comparable to previous studies, which includes only macroscale factors, in Table 4. Figure 1 summarizes the buffer-distance selection results for Steps 1 and 2. In the first row of Figure 1, only the macroscale buffer distance varies, from 200 m to 1600 m on the Y-axis, and cell colors represent marginal R^2, with darker blue indicating lower values. The second row adds microscale features and displays R^2 across macroscale and microscale buffer distances. Because Step 3 separates microscale and visual-quality buffer distances, those results are presented separately in Figure 2 and discussed later.

The findings indicated that 1200m buffer yielded the best fit across most groups, except for Recreational group and Elderly, which peaked at 1400m and 1000m, respectively. The difference

Table 4. Best-fitting macroscale buffer distances and R^2 variation across macroscale buffer distances.

	FS	Util	Rec	Male	Female	Elderly	Inf
Macro index	0.0857*** (5.114)	0.0858*** (5.118)	0.0837** (3.118)	0.0865*** (5.021)	0.0806*** (4.529)	0.0805*** (3.815)	0.0761*** (3.556)
Obs.	29840	27403	2437	13795	16045	5036	5680
Optimal macroscale distance	1200m	1200m	1400m	1200m	1200m	1000m	1200m
Marginal R^2	0.0153	0.0151	0.0191	0.0137	0.0129	0.0119	0.0116
Conditional R^2	0.395	0.393	0.466	0.383	0.411	0.406	0.399
Max difference in R^2 across macroscale buffer distance	0.00598	0.00576	0.00768	0.00588	0.00538	0.00503	0.00269

Note 1. The Full Sample model controlled for trip purpose, visit frequency, age, survey year, gender, and occupation. Each group models excluded the respective variable that defines that group.
Note 2. Group abbreviations: FS: Full Sample, Util: Utilitarian, Rec: Recreational, Inf: Infrequent visitors.
Note 3. Reported in coefficient estimate *** (t-statistic) format; † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

in R^2 across buffer distances (i.e., its sensitivity to buffer distances) was marginal, with the absolute differences between maximum and minimum R^2 per group less than 1 percentage point (p.p.) for all groups. Overall, these results align with most of the past studies which used 1 km~1.5 km buffers for macroscale factors (Adams et al., 2014; Frank et al., 2005; James et al., 2014; Sa and Ardern, 2014).

The coefficient estimates of macroscale index was positive and highly statistically significant and similar across groups. Despite significant coefficient estimates, the absolute values of the marginal R^2 based on macroscale factors was low, in the low to mid-1% range.

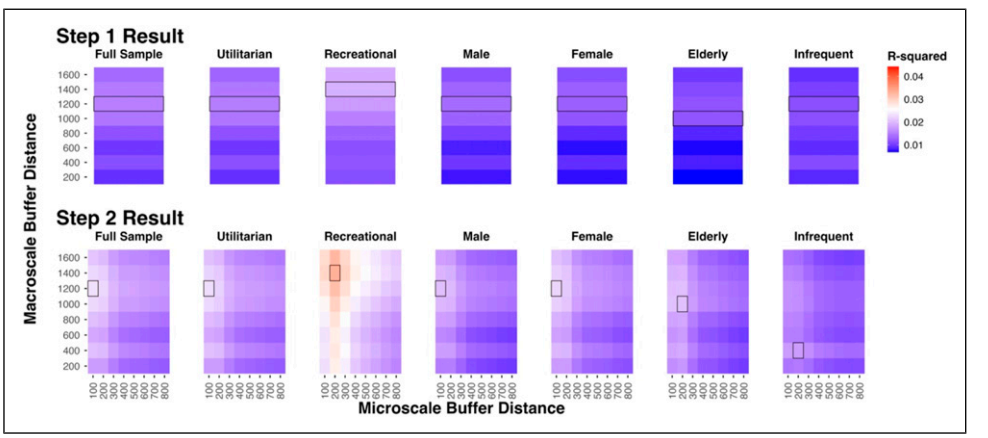


Figure 1. Heatmap of R-squared for Step 1 and Step 2. Groups are represented in columns. Cells with black outline indicate the best-fitting buffer distances. The first row presents Step 1 results: the buffer ranges from 200m to 1600m on the Y-axis and shows R^2 using the color: Darker the blue, the lower the marginal R^2 and brighter the red, the higher the R^2. The second row shows Step 2 results with microscale features added to Step 1. Note that no X-axis exists on the first row because Step 1 does not include microscale features and visual quality measures.

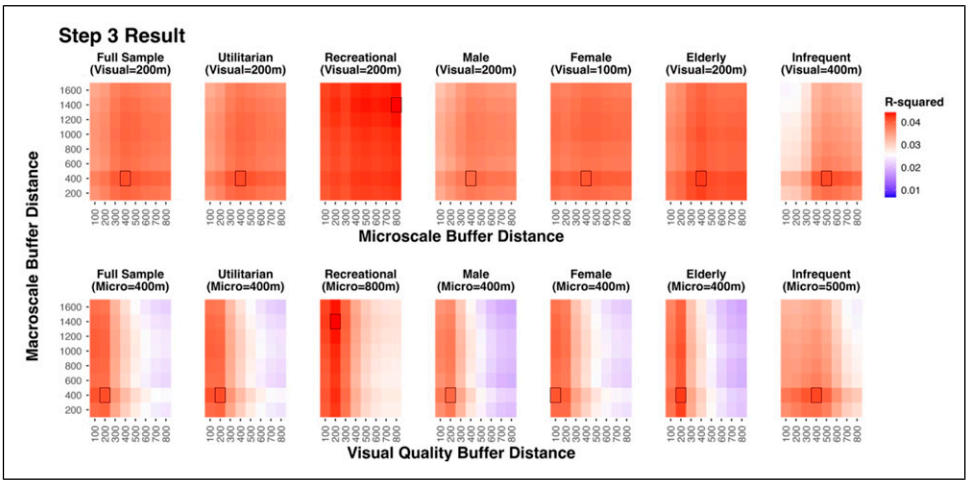


Figure 2. Heatmap of R-squared for Step 3. Note that Step 3 includes three distinct dimensions of buffer distances (i.e., macroscale, microscale, and visual quality buffers), which cannot be displayed on 2D graphics. Therefore, the first and second rows in this figure illustrate the R^2 of the same model with different X-axis: the first row displays how R^2 changes across microscale buffer distances when visual quality buffer distance is fixed at its optimal value; the second row displays changes in R^2 across visual quality buffer distances when microscale buffer distance is fixed at its optimal value.

Step 2: buffer configuration with microscale features included

Step 2 added microscale features—building-to-street ratio and greenness—to the model in the previous step. The resulting best-fitting buffer distances and model fits are presented in the 2nd row of Figure 1 and Table 5). Macroscale index remained statistically significant in all groups with similar, albeit slightly smaller, coefficient estimates and t-statistics. This indicates consistent role of macroscale factors in shaping walking environment satisfaction even after controlling for microscale features.

The regression coefficient estimates of building-to-street ratio is statistically significant at $\alpha = 0.05$ across groups except for Infrequent Visitors, which is significant only at $\alpha = 0.1$. The estimated coefficients are negative across all models, deviating from the expectations based on past findings (Koo et al., 2021). A 1-unit increase in building-to-street ratio is associated with as large as -0.0906 decrease in walking environment satisfaction for Recreational group, and as small as -0.0464 decrease for Infrequent Visitors group. Greenness, on the other hand, is positively associated with walking environment satisfaction for Full Sample, Utilitarian, and Female groups at $\alpha = 0.05$ and Elderly at $\alpha = 0.1$. The magnitude of the coefficients was slightly smaller than building-to-street ratio variable, ranging between 0.0348 and 0.0525.

After adding microscale features, the best-fitting macroscale buffer distances remained identical to Step 1 for all but Infrequent visitors, which showed decreased buffer distances of 400m. The best-fitting buffer distance of microscale features were between 100m and 200m for all groups.

Compared to Step 1, incorporating microscale features exhibited meaningful improvements in R^2. The absolute values of the marginal R^2 reached 2% or higher across all groups except Infrequent Visitors. Recreational walkers' R^2 were particularly high, reaching 3% range.

Table 5. Optimal macroscale and microscale buffer distances and R^2 variation across microscale buffer distances under fixed macroscale settings.

	FS	Util	Rec	Male	Female	Elderly	Inf
Macro index	0.0824*** (4.945)	0.0826*** (4.957)	0.0727** (2.709)	0.0838*** (4.878)	0.0766*** (4.334)	0.0715*** (3.387)	0.0777*** (3.487)
Building-to-street Ratio	−0.0529** (−2.590)	−0.0543** (−2.659)	−0.0906* (−2.557)	−0.0546* (−2.574)	−0.0566** (−2.631)	−0.0551* (−2.049)	−0.0464† (−1.677)
Greenness	0.0441* (2.151)	0.0423* (2.073)	0.0359 (0.996)	0.0348 (1.630)	0.0525* (2.442)	0.0483† (1.799)	0.0402 (1.474)
Obs.	29840	27403	2437	13795	16045	5036	5680
Optimal buffer distance							
Macroscale	1200m	1200m	1400m	1200m	1200m	1000m	400m
Microscale	100m	100m	200m	100m	100m	200m	200m
Marginal R^2	0.0231	0.0229	0.0330	0.0205	0.0223	0.0207	0.0177
Conditional R^2	0.395	0.393	0.469	0.384	0.411	0.406	0.400
Max difference in R^2 across microscale buffer distance	0.0061	0.0060	0.0113	0.0057	0.0068	0.0069	0.0041
Percent change in R^2 from step 1 to 2 (unit: %)	51.00	51.04	73.40	48.97	73.62	74.21	52.76

Note 1. The Full Sample model controlled for trip purpose, visit frequency, age, survey year, gender, and occupation. Each group model excluded the respective variable that defines that group.

Note 2. Group abbreviations: FS: Full Sample, Util: Utilitarian, Rec: Recreational, Inf: Infrequent visitors.

Note 3. Reported in coefficient estimate *** (t-statistic) format; † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 6. Optimal macroscale, microscale, and visual quality buffer distances and R^2 variation across image-based buffer distances under fixed macroscale settings.

	FS	Util	Rec	Male	Female	Elderly	Inf
Macro index	0.0619*** (3.812)	0.0646*** (3.977)	0.0417 (1.481)	0.0601*** (3.592)	0.0576*** (3.330)	0.0493* (2.387)	0.0824*** (3.763)
Building-to-street Ratio	0.0604* (2.401)	0.0564* (2.250)	0.0318 (0.730)	0.0475† (1.831)	0.0411 (1.629)	0.0509 (1.599)	0.0912** (2.605)
Greenness	−0.0369 (−1.507)	−0.0409† (−1.678)	−0.0347 (−0.776)	−0.0563* (−2.225)	−0.0033 (−0.129)	−0.049 (−1.583)	−0.0602† (−1.889)
Safety	0.2732*** (9.717)	0.2754*** (9.730)	0.2218*** (5.454)	0.2808*** (9.429)	0.2257*** (9.097)	0.2874*** (8.085)	0.3298*** (7.650)
Liveliness	−0.2312*** (−7.881)	−0.2329*** (−7.897)	−0.1796*** (−4.517)	−0.2376*** (−7.704)	−0.1890*** (−7.434)	−0.2452*** (−6.745)	−0.3128*** (−6.558)
Obs.	29840	27403	2437	13795	16045	5036	5680
Optimal buffer distance							
Macroscale	400m	400m	1400m	400m	400m	400m	400m
Microscale	400m	400m	800m	400m	400m	400m	500m
Visual quality	200m	200m	200m	200m	100m	200m	400m
Marginal R^2	0.041	0.041	0.044	0.040	0.041	0.042	0.042
Conditional R^2	0.396	0.393	0.471	0.384	0.412	0.407	0.401
Max difference in R^2 across:							
Microscale buffers	0.0040	0.0041	0.0021	0.0045	0.0011	0.0041	0.0090
Visual quality buffers	0.0162	0.0164	0.0158	0.0179	0.0164	0.0212	0.0110
Percent change in R^2 from step 2 to 3 (unit: %)	78.41	81.06	33.98	93.07	82.77	105.02	136.03
Percent change in R^2 from step 1 to 3 (unit: %)	169.40	173.47	132.33	187.63	217.34	257.17	260.56

Note 1. The Full Sample model controlled for trip purpose, visit frequency, age, survey year, gender, and occupation. Each group model excluded the respective variable that defines that group.
Note 2. Group abbreviations: FS: Full Sample, Util: Utilitarian, Rec: Recreational, Inf: Infrequent visitors.
Note 3. Reported in coefficient estimate *** (t-statistic) format; † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Step 3: buffer configuration for full model

As the final step, visual quality measures—safety and liveliness—were added to the models (See Table 6 and Figure 2). Because Step 3 evaluates three buffer dimensions, a single 2D heatmap would require one dimension to be omitted or collapsed. Figure 2 shows two 2D slices of the three-dimensional buffer search space: one varying macroscale and microscale buffers with visual quality fixed at its optimum, and the other varying macroscale and visual-quality buffers with microscale fixed at its optimum. With all variables entered, the highest variance inflation factor (VIF) was 2.25 for liveliness.

The inclusion of visual quality measures changed the coefficient estimates of building-to-street ratio from negative to positive, but tended to weaken the association of microscale features. For Recreational group, neither macroscale nor microscale features remained statistically significant; only visual quality measures were significant predictors.

Safety was statistically significant with large positive coefficient estimates and t-statistics. A 1-unit increase in safety was associated with as large as 0.3298 increase in walking environment satisfaction for Infrequent Visitors and as small as 0.2218 increase for Recreational groups. In general, the magnitude of the coefficient estimates of safety were more than four to five times that of macroscale index.

Liveliness was strongly, but inversely, associated with walking environment satisfaction for all groups. A 1-unit increase in liveliness was associated with 0.3128 decrease in walking environment satisfaction for Infrequent Visitors. This negative association is also an unexpected finding, contrasting with the literature, which tends to regard lively urban streets as inviting for pedestrian activities (Jacobs, 1961; Sung et al., 2015).

In general, adding the visual quality measures increased the best-fitting microscale buffer distances from 100-200m to 400-500m for most groups. In contrast, macroscale buffer distances decreased from over 1000m down to 400m for most groups, converging with the buffer distances of microscale features. Recreational group was an exception in this regard: The optimum buffer distances of macroscale and microscale were at 1400m and 800m, respectively.

Importantly, visual quality measures were sensitive to buffer distances, resulting in R^2 differences as large as 2.12 percentage points across visual quality buffer distances while holding macroscale and microscale buffer distances constant at their optimum. This difference is comparable or even larger than the absolute R^2 values themselves in Step 1 and 2. When the best-fitting distance is used, adding visual quality measures offered substantial increase to R^2, elevating them to above 4% for all groups. Compared to the models in Step 2, this represents a 78.4% increase in R^2 for Full Samples and up to 136.0% for Infrequent Visitors.

Discussion

This study investigated the spatial reach of GIS-based macroscale factors and image-based microscale and visual quality measures by identifying the buffer distances that provide the strongest explanatory power for walking environment satisfaction. This study also examined how the best-fitting buffer distances vary by trip purposes and pedestrian groups. Our results demonstrate that the best-fitting buffer distances for macroscale factors and visual information for pedestrian environment can vary substantially and that spatial reach of visual quality measures tend to be much shorter than macroscale or microscale measures.

Step 1 of the previous section showed that macroscale index alone provided poor model fits across groups, with the optimum buffer distances between 1000 and 1400 m. Step 2 showed that microscale features provided improvements to the model fits. The best-fitting buffer distances of

macroscale factors in Step 2 remained above 1 km for most groups while the best microscale buffer distances were uniformly 200m or less across all groups.

In Step 3, visual quality measures introduced the most substantial improvements in R^2. Macroscale buffer distances decreased significantly from above 1 km down to 400m for all but Recreational group. This decrease may be due to the covariance structure among macroscale index and visual quality measures. As its buffer distance increases, macroscale index tends to be more strongly correlated with safety (the highest correlation of $r = 0.45$ occurred when macroscale and visual quality buffer distances were 1600m and 800m, respectively) and is least correlated when macroscale buffer distance is 400m (as low as $r = -0.01$; See Appendix C for details). From a variance-partitioning perspective, the part of the variance previously explained by macroscale index from large buffers in Step 1 may be now explained by correlated visual quality measures in Step 3, making macroscale index from large buffers that much redundant.

The finding that microscale features and visual quality measures can have different optimal buffer distances is also important. Building-to-street ratio and greenness are measured from street-view images, but they may primarily capture urban morphological characteristics. Urban morphology, particularly building-to-street ratio, is largely shaped by zoning regulation, which governs the building height and the road width based on anticipated density and use pattern of the area. It is therefore plausible that building-to-street ratio and greenness can remain spatially consistent within the same zone for hundreds of meters. In contrast, safety and liveliness may be shaped by non-morphological characteristics such as specific street objects, lighting, maintenance quality, crowdedness, or the density of activities. These qualitative characteristics can vary street by street as they are not governed systematically by zoning regulations and can require close-range observation to be accurately assessed (Gehl, 2013). Evaluative judgments of safety and liveliness can be more sensitive to the immediate environment in close proximity to the survey location resulting in a shorter optimal buffer distance than morphological indicators such as greenness or building-to-street ratio.

Macroscale, microscale, and visual quality measures showed varying levels of sensitivity to buffer distances. Macroscale and microscale measures in Steps 1 and 2 showed minor differences in R^2 across buffer distances. This finding aligns with the past studies that reported insensitivity of macroscale factors to buffer distances (e.g., Villanueva et al., 2014) and partly explains the lack of consensus in earlier literature regarding appropriate buffer distances. However, visual quality measures can be sensitive to buffer distance: The differences in R^2 can be as large as 2.12 percentage points depending on buffer distances. This difference is larger than the highest R^2 of Step 1 models. This suggests the importance of selecting the right buffer distances—200 m in most cases and underscores the importance of visual context of immediate surrounding in determining the walking environment satisfaction.

Although the absolute marginal R^2 values are modest, the relative improvements across model steps are substantial. The addition of image-based indicators in Steps 2 and 3 increased marginal R^2 by 74.2%-260.6% relative to the macroscale-only model, depending on the subgroup. This pattern is substantively important because the macroscale index used in Step 1 follows one of the widely cited walkability indices (Frank et al., 2005); the added explanatory gain therefore indicates that image-based indicators capture meaningful variation in walking environment satisfaction beyond what macroscale built-environment factors alone explain, particularly so when the right buffer distances are used for measurement.

The modest R^2 values are also consistent with prior studies of walking-related outcomes using built-environment characteristics. For example, Lovasi et al. (2008) reported that individual built-environment characteristics explained less than 0.6% of the variance in walking-for-exercise time, and their multivariate model explained 5.6% of the variation in the training set and less than 1% in the validation set. Grasser et al. (2016) reported R^2 values of 7%-13% for neighborhood

satisfaction, Lee et al. (2018) reported R^2 values of 1%-10% using multiple Seoul data sources, Kim and Jin (2023) found that accessibility variables alone explained approximately 3% of walkability variance in Daegu, South Korea, although including perception variables and individual characteristics from the survey increased R^2 to over 30%.

We suspect the low absolute explanatory power likely reflects inherent characteristics of the outcome variable rather than the irrelevance of the predictors. Walking environment satisfaction is a perceptual evaluation that may be shaped by unobserved individual-level factors, including mood, prior expectations, personal preferences, and weather conditions at the time of the survey. In particular, among these individual-level factors, walking-related preferences and attitudes can strongly shape how individuals evaluate their walking environment (e.g., Van der Vlugt et al., 2025). However, due to data limitations, this study could include only a limited set of individual characteristics, such as age, and could not account for more nuanced personal factors. The restricted variance of a bounded ordinal scale may further constrain achievable R^2.

There are two notable findings regarding the regression coefficient estimates. First, while most microscale features in Step 2 were statistically significant in most cases, which aligns with past findings, visual quality measures in Step 3 rendered greenness insignificant for all but Male group. This may be partly due to the correlations between visual quality and microscale features. The correlation coefficient between liveliness and building-to-street ratio is 0.42 and −0.28 for greenness when measured at their optimal buffer distances in Step 3 (See Appendix E). Safety and greenness also covary with correlation coefficient of 0.31. The second notable finding is that liveliness showed negative coefficient estimates, deviating from expectations. This might be due to the unique characteristics of Seoul, Korea. Extreme population densities past a certain threshold can have significantly reduced association with walking behavior (Eom and Cho, 2015) or can even have inverse associations. Cerin et al. (2022) showed that population density has an inverted-U shape relationship with walking probability which peaks when the population density is around 13,000∼14,000 persons/km^2. The population density of Seoul is 17,000 persons/km^2 (Joo et al., 2025), exceeding this threshold. Streetscapes bustling with people and their activity (i.e., high liveliness) are likely to be of high density and may be hard to navigate due to limited spaces (See Figure 3). Such streets may be more pleasant for walking up to a degree, but when streets get too busy, they may be over-stimulating and cause fatigue, casting negative impacts on pedestrians (Eom and Cho, 2015).

It is important to note that the coefficient estimates of safety, as measured using the Place Pulse 2.0 dataset, may reflect more than just safety per se. The data collection process of Place Pulse 2.0 did not clearly instruct the labelers to exclude from consideration such features as cleanliness, design quality, maintenance, lighting, and perceived social character, all of which can influence walking environment satisfaction. The safety scores, therefore, likely reflect various visual cues that people take into consideration when judging safety of a scene (Ki et al., 2026; see Figure 3). This interpretation is consistent with the finding that the coefficient estimate of visual safety was greater for infrequent visitors: Pedestrians in an unfamiliar area tend to rely more heavily on visual cues to judge whether a place feels safe and comfortable to walk in (Kang et al., 2023; Zhang and Bandara, 2024). In this sense, visual safety can be understood as a primary perceptual gateway to walking satisfaction for those without prior experience of a place. Future research is needed to rigorously partition the true visual safety and liveliness from other bundled visual characteristics.

In general, the Full Sample, Male, and Female showed similar best-fitting buffer distances, regression coefficients, and t-statistics across groups in Steps 1 through 3 despite the widely varying sample sizes across them. However, Recreational Walkers and Infrequent Visitors were two groups that showed noticeably different patterns in the buffer distances and R^2. For Recreational Walker, microscale features in Step 2 offered a greater jump in the model fit compared to other groups. When visual quality measures enter the models, the optimal macroscale buffer distances of other groups reduce to

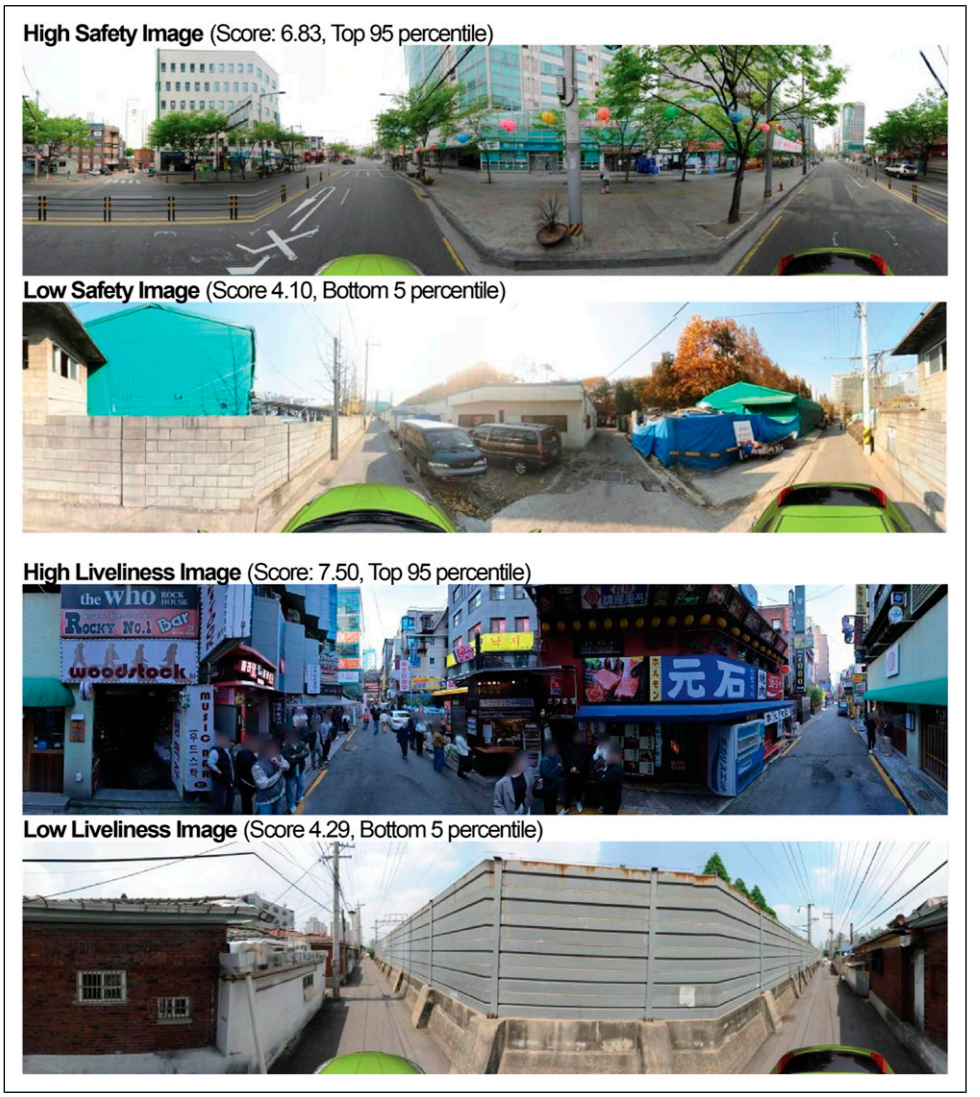


Figure 3. Examples of streetscapes with different levels of safety and liveliness.

400m while that of Recreational group remains at 1400m. The larger buffer distances for recreational walking compared to other trip purposes is consistent with past findings (Sugiyama et al., 2019).

Elderly group did not exhibit substantial differences in optimal buffer distances compared to the Full Sample group. This is an unexpected result, contradicting some of the past findings such as Stafford & Baldwin (2018), which argued that older adults—due to physical ability constraints—may require smaller buffer distances than those applied to the general population. We speculate sampling bias as one of the possible reasons. The Seoul Survey was an intercept survey, and older adults who have difficulties in mobility may be less likely to be out in the survey locations, and those that responded to the survey may be biased towards those who can walk by themselves, making them similar to the general population.

Several policy implications can be derived from the findings of this study. First, planning interventions aimed at offering a more satisfying walking environment at a given location can consider

different spatial extents depending on whether the focus is on macroscale, microscale, or visual quality features. Policies centered around macroscale factors, such as land use mix, residential density, and intersection density, may benefit from considering larger spatial extents (i.e., 1,000-1,400m in Steps 1 and 2). In contrast, interventions focused on improving visual quality of streetscapes may use more location- or route-specific approaches, given that its associations with walking environment satisfaction were strongest when measured within relatively short distance ranges.

Another policy implication pertains to the differences in the associated factors and their buffer distances across trip purposes and user groups. This finding provides a basis for considering context- and target-specific strategies rather than one-size-fits-all approaches. For example, planners working on improving the walking environment in recreational areas may place greater consideration on streetscape elements related to the visual quality of immediate vicinities than on other broader-area characteristics such as density and land use mix. Several studies have identified streetscape features associated with higher perceived safety, including greater visibility of tall trees and other non-view-obstructing vegetation, depending on land use types (Li et al., 2015), stronger streetscape enclosure and a greater number of buildings along the block (Harvey et al., 2015), and street-facing windows that support natural surveillance (Navarrete-Hernandez et al., 2021). Another example pertains to Infrequent visitors who were the only subgroup for which macroscale index, microscale, and visual quality were all statistically significant at least at 90% confidence level, and they generally exhibited the largest coefficient magnitudes. These findings suggest that, for places attracting large numbers of visitors unfamiliar with the area, such as tourist districts, major transfer centers, and regional commercial centers, planners may consider multiple dimensions of the built environment simultaneously, including accessibility (e.g., greater land use mix and better connected street networks with greater intersection density), streetscape morphology, and visual quality rather than focusing on only one dimension of the built environment.

Lastly, the findings suggest that image-based visual quality can be incorporated into walkability assessment frameworks. Existing walkability indices typically rely on GIS-derived macroscale measures such as connectivity, land-use mix, and density. In our analyses, however, adding image-based visual quality measures substantially improved model fit, increasing the marginal R^2 by up to 136% even compared to models including macroscale and microscale variables. These improvements were achieved when visual quality was measured using appropriate short-range buffers (typically $\leq$200 m). Given the important role of walkability indices in promoting public awareness and informing policy decisions, this incorporation may help draw greater attention to streetscape design and how safe and pleasant streets look, thereby providing additional information for planning and policy decisions.

There are some important limitations of this study, which provides future research directions. First, we assumed that a person would evaluate the walking environment of the surveyed location based on different sets of considerations depending on their trip purposes. This may not always be true, particularly when the survey respondents are familiar with the survey location and therefore have already accumulated experience of that location.

Second, this study chose to use Euclidean buffer over network buffer to reflect the large apartment complexes in Seoul where road geometry is missing in GIS data. Network buffers, which rely on road geometry, would ignore the substantial number of residential units in these apartment complexes in residential density and land use mix calculation. However, several studies reported network buffers as being better at accurately capturing the walkable environment for various walking and physical activity-related outcome measures (James et al., 2014; Li et al., 2022; Oliver et al., 2007). According to a simulation study, Euclidean buffers underestimate the association between built environment on transport walking (Li et al., 2022). Future studies should benefit from using network buffers or sausage buffers over Euclidean buffers.

Third, this study used a decade-old walking environment satisfaction survey. This decision was made due to the lack of alternative and more recent datasets. The City of Seoul discontinued the release of walking environment satisfaction survey a decade ago, and to our best knowledge, no other alternative public datasets have been released. Despite this limitation, the use of this dataset offers valuable insights to the audiences because (1) individual-specific characteristics related to neighborhood or walking environment preference are often thought to be time-constant factors (Knuiman et al., 2014), (2) longitudinal studies have found consistent associations between the built environment and outcomes over time (Knuiman et al., 2014; Li et al., 2022; McCormack et al., 2021), and (3) several systematic review papers have offered evidence supporting that the relationship between the built environment and walking is consistent over time (Bandara et al., 2025; Kärmeniemi et al., 2018; Public Health Agency of Canada, 2018). Regardless, future studies can benefit from using a more recent survey to evaluate the findings of this study.

Fourth, this study is subject to the inherent limitations of the intercept survey design. Although the survey was conducted by trained interviewers and included both weekday and weekend responses, the sample may overrepresent individuals who were present at the selected locations during the survey period and who were more willing to engage with interviewers. Although the sample shows reasonable similarity to the demographic profile of Seoul, unobserved individual characteristics may still introduce residual biases. This limitation may be particularly relevant because individuals who avoid certain walking environments due to poor perceived safety, comfort, or accessibility are not captured. Therefore, the measured satisfaction may better represent active users of the surveyed locations than the broader population of potential pedestrians. The population representativeness of the findings remains uncertain, and the policy implications should be interpreted with caution.

Finally, because the dependent variable is walking environment satisfaction rather than observed walking behavior, all findings and policy implications derived from the findings should be interpreted as applying to perceived quality of the walking environment. Generalizing these buffer distances to studies predicting actual walking behavior requires further investigation, because walking behavior may be shaped by trip necessity, time constraints, and alternative mode availability in addition to the immediate visual environment.

Conclusion

The findings of this study contribute to the literature in several important ways. First, this study empirically investigated the reach of visual information, and therefore buffer distances, of image-based measures of microscale and visual quality. Second, the group analysis revealed that optimal buffer distances and model improvements differ significantly between utilitarian versus recreational walking. Third, this study reports empirical insights in the context of a city with a high-density, highlighting that livelier streetscapes are inversely associated with walking environment satisfaction. The results of this study can help better understand the true spatial extent in which accessibility and image-based measures of walkability should be operationalized, improving both the precision of research and the relevance of planning and policy interventions.

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Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This research was supported by the Yonsei University Research Fund of 2025-22-0145.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

Data will be made openly available on public data repository.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT and Gemini in order to improve English grammar and language clarity. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Supplemental material

Supplemental material for this article is available online.

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